CS 470 Project Two Conference Presentation: Cloud Development

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Introduction

Residing in Oklahoma, raised in Bethel, New York.

Who I am: Developer for QnA, a startup company.

Project: Finished developing a tack web application.

A good problem to have!!: Increased userbase created growing pains with our local database setup and management.

Solution implemented: Go serverless!



Containerization

Docker Desktop or Docker CLI for Linux can enable quick development and rehosting capabilities.

These tools allo developers to package frontend and backend components into portable runtime entities.



Orchestration

- Coordinating multi-container applications
  - Multi-Service Definition
  - Automated networking
  - Simplified development



The Serverless Cloud

- What is “serverless” and what are its advantages?
  - Reduces overhead
  - Automatic Scaling
  - Cost efficiency
- What is S3 storage and how does it comp[illegible]ocal storage?
  - Local store is limited to one machine while S3 is global.
  - Maintenance is high with local and low on global S3 buckets.
  - Local has fixed capacity limit and S3 is unlimited.
  - The reliability for local storage is low and with S3 it is highly durable and carries redundancies for sensitive data storage



The Serverless Cloud

API & Lambda

- Serverless API Benefits
  - Zero idle server costs, high availability, and d scaling under peak loads.
- Lambda Execution Logic
  - Event-driven functions execute discrete request handlers on demand.
- Frontend Integration
  - Configure API Gateway routes, provision deployment scripts, and connect client SDKs.



The Serverless Cloud

Database Architectures

MongoDB vs. DynamoDB

- Data-Model Differences
  - MongoDB (Document Model): Highly flexible schema allowing nested documents, dynamic fields, and rich querying across hierarchical data structures.
  - DynamoDB (Key-Value & Document Model): Strict partition/sort key design requiring deliberate access-pattern planning, single-table design principles, and predictable scaling limits.

DynamoDB Queries

Single Item Lookups

Fast retrieval via primary partition and sort keys.

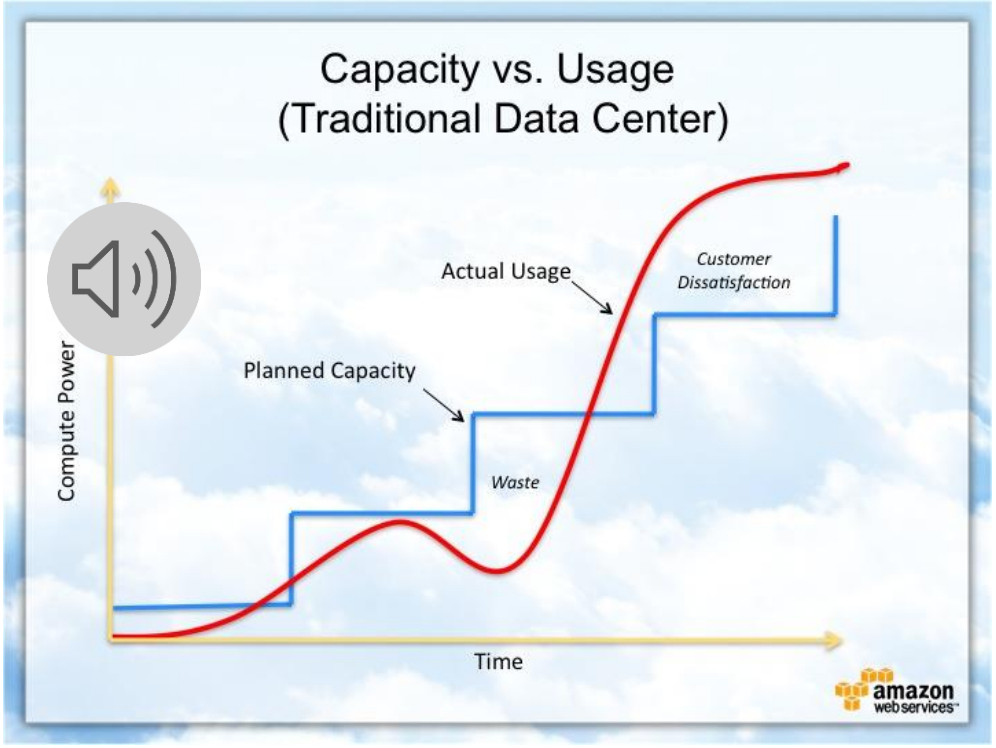
```
const dynamoDbClient = new DynamoDBClient();
const docClient = DynamoDBDocument.from(dynamoDbClient);
...more code
let scanResult = await docClient.send(new mand(params));
let data = scanResult.Items ? scanResult.It] : null;
```

Event JSON test with filters.

```
{"resource": "/Question",
"httpMethod": "GET",
"queryStringParameters": {
"filter": "{\"include\":{\"relation\":\"answers\"},\"where\":{\"categorySlug\":\"angular\"}}" }
```

Cloud-Based Development Principles

- Elasticity
- Pay-for-use model





Securing Your Cloud App

Access

- Apply principles of least privilege (IAM policies)
- Keep S3 buckets private.
- Validate inputs

Policies

- A policy is a document written in JS[illegible]at defines wh[illegible]s are allowed or d[illegible]and on which resources.
- A role is an identity that has specific permissions.

API Security

- Secure the connection between API Gateway and Lambda by using IAM policies, authorization controls, and least-privilege access to ensure only trusted requests can invoke functions.
- Protect communication between Lambda and the database by using private networking, encrypted connections, and secure credential management through secrets storage.
- Secure S3 buckets by enforcing encryption, restricting public access, and applying role-based permissions to control data access.

CONCLUSION

- Migrating from a traditional full-stack to serverless our workload is reduce, allowing rapid development, a focus on innovation, and it creates direct business value.
- Leveraging cloud-native solutions such as serverless functions, managed databases, and automated deployment provides scalability, reliability, and flexibility needed to transform concepts into production-ready solutions.
- Aligning architectural decisions with business objectives ensures lower operational costs, faster deployment cycles, and improved organizational agility, ultimately enabling ideas to become measurable and sustainable business outcomes.

Thank you for your time.